

Nephrology Clinical Assessment Report

Chronic Kidney Disease Screening · Confidential Clinical Document

Patient Information

Patient Name	Margaret T.	Patient ID	CKD-2026-002
Date of Report	April 29, 2026	Exam Type	Nephrology Screening — CKD Assessment
Age / Gender	65 years · Female	Referring Department	Internal Medicine
Hypertension	Yes — currently managed	Diabetes Mellitus	Yes — confirmed
Coronary Artery Disease	Yes — confirmed	Informant	Patient / Medical Record

Laboratory Findings — Quantitative Markers

Parameter	Result	Reference Range	Status	Clinical Note
Blood Pressure	160 mmHg	<120 mmHg	HIGH	Stage 2 hypertension — accelerates glomerular damage
Specific Gravity	1.010	1.010–1.030	VERY LOW	Impaired urinary concentration — tubular dysfunction
Albumin (urine)	2+	Negative / Trace	HIGH	Significant proteinuria — glomerular membrane compromise
Blood Sugar (urine)	Positive	Negative	ABNORMAL	Glycosuria — indicative of poorly controlled diabetes
BUN	90 mg/dL	7–20 mg/dL	VERY HIGH	Severe azotemia — marked reduction in renal clearance
Serum Creatinine	3.5 mg/dL	0.5–1.1 mg/dL	VERY HIGH	eGFR critically reduced — consistent with CKD Stage 4–5
Sodium	132 mEq/L	136–145 mEq/L	LOW	Hyponatremia — impaired tubular sodium reabsorption
Potassium	6.5 mEq/L	3.5–5.0 mEq/L	HIGH	Hyperkalemia — risk of fatal cardiac arrhythmia
Hemoglobin	8.0 g/dL	12.0–16.0 g/dL	VERY LOW	Severe renal anemia — reduced erythropoietin production
PCV	25%	36–48%	LOW	Hematocrit critically reduced — confirms anemia severity

Laboratory Findings — Qualitative & Clinical Markers

Marker	Finding	Status	Clinical Significance
RBC Classification	Abnormal	FLAGGED	Dysmorphic RBCs suggest glomerulonephritis or tubular bleeding
WBC Classification	Abnormal	FLAGGED	Elevated urinary WBCs indicate active renal inflammation
Pus Cells	Abnormal	FLAGGED	Pyuria — active urinary tract infection or tubulointerstitial nephritis
Pus Cell Clumps	Present	FLAGGED	Clumped pus cells — strong indicator of upper UTI or renal abscess
Bacteria	Present	FLAGGED	Confirmed bacteriuria — urinary tract infection requiring treatment
Hypertension	Yes	FLAGGED	Major bidirectional CKD risk factor — driving glomerular hypertension
Diabetes Mellitus	Yes	FLAGGED	Leading cause of CKD globally — diabetic nephropathy likely etiology
Coronary Artery Disease	Yes	FLAGGED	Severe cardiovascular complication — CKD and CAD mutually worsen prognosis
Appetite	Poor	FLAGGED	Uremic anorexia — hallmark symptom of advanced renal failure
Pedal Edema	Present	FLAGGED	Bilateral fluid retention — hallmark sign of nephrotic syndrome / CKD
Anemia	Confirmed	FLAGGED	Renal anemia from EPO deficiency — clinically severe at current Hb level

All 10 quantitative markers deviate from reference ranges. All 11 qualitative markers flagged as abnormal or present. Complete convergence of findings across renal, hematological, infectious, and cardiovascular domains.

Clinical Observations

Margaret T. is a 65-year-old female with a confirmed triad of hypertension, diabetes mellitus, and coronary artery disease — representing three of the most potent independent risk factors for chronic kidney disease. She presents with poor appetite, bilateral pedal edema, and confirmed clinical anemia. Urinalysis reveals significant proteinuria (2+), glycosuria, bacteriuria, pyuria, and the presence of pus cell clumps, indicating active urinary tract infection superimposed on advanced structural renal disease. Both RBC and WBC classifications are abnormal. Serum markers confirm severe azotemia (BUN 90 mg/dL), critically elevated creatinine (3.5 mg/dL), hyperkalemia (K⁺ 6.5 mEq/L), and hyponatremia (Na⁺ 132 mEq/L). Hematological indices confirm severe anemia with hemoglobin at 8.0 g/dL and PCV at 25%.

Clinical Impression

The clinical, laboratory, and symptomatic profile of Margaret T. is consistent with end-stage Chronic Kidney Disease (CKD Stage 4–5), most likely secondary to diabetic nephropathy and hypertensive nephrosclerosis. The

breadth and severity of abnormalities across renal function, urinalysis, hematology, and electrolyte balance — combined with the presence of active urinary infection, pedal edema, and cardiovascular comorbidity — constitutes one of the most complete and severe CKD presentations. The elevated potassium level (6.5 mEq/L) poses an immediate risk of life-threatening cardiac arrhythmia and requires urgent management. This report does not substitute for a formal clinical diagnosis.

Recommendations

1. **Emergency electrolyte management:** Hyperkalemia (K⁺ 6.5) requires immediate treatment — calcium gluconate, insulin-dextrose, or kayexalate as indicated; continuous cardiac monitoring.
2. Urgent nephrology referral for formal CKD staging, eGFR calculation, and renal replacement therapy planning (hemodialysis or peritoneal dialysis).
3. **Antibiotic therapy:** Treat confirmed urinary tract infection with culture-guided antibiotics; upper UTI / pyelonephritis protocol given pus cell clumps.
4. Erythropoiesis-stimulating agent (ESA) therapy and iron status evaluation for renal anemia (Hb 8.0 g/dL); consider transfusion threshold assessment.
5. Strict glycemic control and antihypertensive optimization — target BP <130/80 mmHg; ACE inhibitor or ARB preferred for renoprotection in diabetic nephropathy.
6. Dietary nephrology consultation: strict potassium and phosphate restriction, sodium and fluid management, protein-controlled diet.
7. Cardiology co-management given confirmed CAD and the cardiovascular risk amplification of CKD Stage 4–5 with severe anemia.

Examining Clinician — Nephrology

Date: April 29, 2026